

LungTIME-C01 Randomization Forensics Audit

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1 Scope

This report summarizes randomization-forensics signals from baseline characteristics in:

- s41591-025-04181-w.pdf
- 41591_2025_4181_MOESM1_ESM.pdf

Inputs are generated by:

- scripts/extract_randomization_table1.py
- scripts/build_randomization_inputs.py
- scripts/run_randomization_forensics.R

2 Load Inputs

```
repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
input_dir <- file.path(repo_root, "reports", "randomization", "lungtime")

required_files <- c(
  "row_level_results.csv",
  "pooled_pvalues.csv",
  "simdistr_variable_pvalues.csv",
  "simdistr_combined_pvalues.csv"
)
missing_files <- required_files[!file.exists(file.path(input_dir, required_files))]
if (length(missing_files) > 0) {
  stop(
    "Missing expected analysis outputs: ",
    paste(missing_files, collapse = ", "),
    ". Run scripts/run_randomization_forensics.R first."
  )
}

row_level <- read.csv(file.path(input_dir, "row_level_results.csv"), check.names = FALSE)
pooled <- read.csv(file.path(input_dir, "pooled_pvalues.csv"), check.names = FALSE)
sim_var <- read.csv(file.path(input_dir, "simdistr_variable_pvalues.csv"), check.names = FALSE)
sim_combined <- read.csv(file.path(input_dir, "simdistr_combined_pvalues.csv"), check.names = FALSE)

ascii_text <- function(x) {
  x <- gsub(" ", ">=", x, fixed = TRUE)
  x <- gsub("-", "-", x, fixed = TRUE)
  x
}

row_level$variable <- ascii_text(row_level$variable)
row_level$level <- ascii_text(row_level$level)
sim_var$variable <- ascii_text(sim_var$variable)
sim_var$level <- ascii_text(sim_var$level)
```

3 Trial Metadata Snapshot

```
meta_path <- file.path(repo_root, "data", "processed", "randomization", "lungtime", "randomi
if (file.exists(meta_path)) {
  metadata <- read.csv(meta_path, check.names = FALSE)
  knitr::kable(metadata)
} else {
  cat("Metadata file not found at:", meta_path)
}
```

		al-		alloca-			sourcespage_page_con-		
arm	amlab	la-	randomiza-	ceal-	strat-		source	page	page
trial_id	bel	n_a	bio	tion_method	ment	tion	blinding	port	col
lung-	early	late	051051:1	computer-	sequen-	none	open-label	9	6
time_c01_s41591_025_0418				generated	tially	(with-	for		31
				random	num-	out	treatment		
				number	bered,	strat-	assignment;		
				table by in-	opaque,	ifica-	BIRC		
				dependent	sealed	tion)	blinded for		
				statistician	en-		endpoint		
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4 Pooled Summary Metrics

4.1 Interpretation Key (Heathers-aligned)

The metrics below are randomization-forensics checks on baseline Table 1 distributions, aligned with the screening logic discussed in Heathers’ overview of STALT/CSF-style methods:

- `n_rows_reported`: number of baseline rows with manuscript-reported p-values.
- `n_rows_recalc`: number of baseline rows with recalculated row-level p-values.
- `stouffer_reported`: one-sided Stouffer combination of reported row p-values.
- `stouffer_recalc`: one-sided Stouffer combination of recalculated row p-values.
- `fisher_reported`: Fisher combination of reported row p-values.
- `fisher_recalc`: Fisher combination of recalculated row p-values.
- `simdistr_combined_pvalue`: simulation-based combined p-value from `simdistr`.

Interpretation rule of thumb:

- Smaller combined p-values indicate less compatibility with expected random baseline balance.
- Larger combined p-values indicate no low-tail signal for imbalance under these checks.
- No single metric is dispositive; methods are screening tools and should be interpreted with design/context checks from protocol and data provenance.

```
knitr::kable(pooled, digits = 6)
```

trial_id	n_rows_reported	n_rows_recalc	stouffer_reported	stouffer_recalc	fisher_reported	fisher_recalc
lung-time_c01_s41591_025_04181	17	17	0.999998	1	0.947728	0.859243

```
knitr::kable(sim_combined, digits = 6)
```

simdistr_combined_pvalue	trial_name
0.253716	lungtime_c01_s41591_025_04181

```
n_rows <- nrow(row_level)
n_flagged_delta <- sum(row_level$flagged_p_delta_0_05, na.rm = TRUE)
cat("Rows analyzed:", n_rows, "\n")
```

Rows analyzed: 17

```
cat("Rows with |reported p - recalculated p| >= 0.05:", n_flagged_delta, "\n")
```

Rows with |reported p - recalculated p| >= 0.05: 9

5 Row-Level Signals

5.1 Lowest recalculated p-values

```
row_by_p <- row_level[order(row_level$row_chisq_p, na.last = TRUE), ]
knitr::kable(head(row_by_p[, c("variable", "level", "row_chisq_p", "reported_p")], 10), digits = 5)
```

	variable	level	row_chisq_p	reported_p
13	Tumor burden	>=80 mm	0.07232	0.097
5	Stage	IIIC	0.22298	0.120
7	Liver metastasis	Yes	0.29814	0.300
14	LDH	1 ULN	0.35443	0.440
3	ECOG PS	0	0.37978	0.380
8	PD-L1 TPS	<1%	0.40139	0.862
17	LIPI score	High risk	0.44787	0.673
15	LIPI score	Low risk	0.48504	0.673
10	PD-L1 TPS	>=50%	0.51096	0.862
6	Brain metastasis	Yes	0.54847	0.550

5.2 Largest reported vs recalculated p-value gaps

```
row_by_delta <- row_level[order(-row_level$abs_p_delta, na.last = TRUE), ]
knitr::kable(
  head(row_by_delta[, c("variable", "level", "reported_p", "row_chisq_p", "abs_p_delta")], 10),
  digits = 5
)
```

	variable	level	reported_p	row_chisq_p	abs_p_delta
8	PD-L1 TPS	<1%	0.862	0.40139	0.46061
10	PD-L1 TPS	>=50%	0.862	0.51096	0.35104
17	LIPI score	High risk	0.673	0.44787	0.22513
15	LIPI score	Low risk	0.673	0.48504	0.18796
11	PD-L1 TPS	Unknown	0.862	1.00000	0.13800
5	Stage	IIIC	0.120	0.22298	0.10298
9	PD-L1 TPS	1-49%	0.862	0.75998	0.10202
16	LIPI score	Medium risk	0.673	0.77124	0.09824
14	LDH	1 ULN	0.440	0.35443	0.08557
13	Tumor burden	>=80 mm	0.097	0.07232	0.02468

6 simdistr Variable-Level Signals

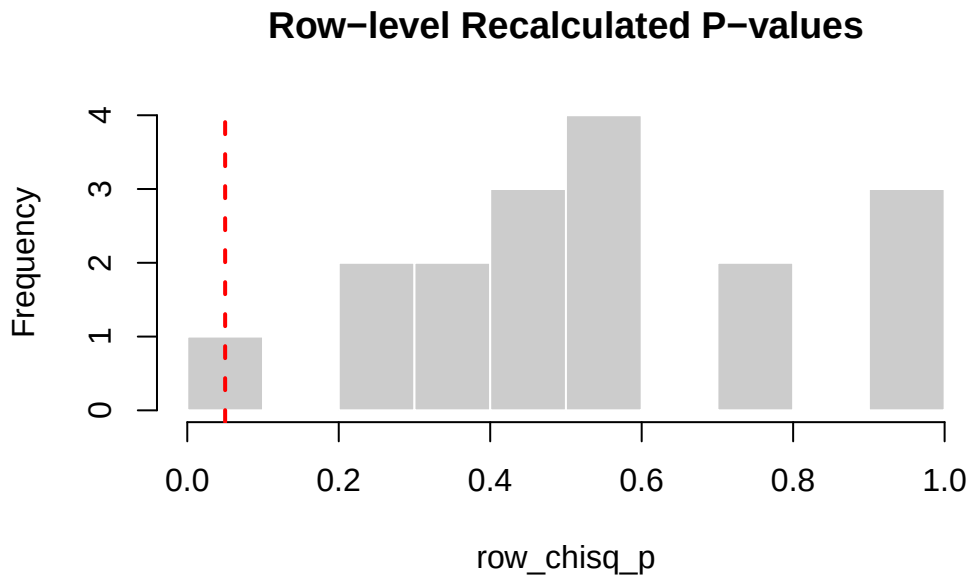
```
sim_by_p <- sim_var[order(sim_var$simdistr_pvalue, na.last = TRUE), ]  
knitr::kable(head(sim_by_p[, c("variable", "level", "simdistr_pvalue")], 15), digits = 5)
```

	variable	level	simdistr_pvalue
12	ICIs	Sintilimab	0.02540
1	Sex	Male	0.03855
11	PD-L1 TPS	Unknown	0.05265
16	LIPI score	Medium risk	0.28040
9	PD-L1 TPS	1-49%	0.29520
2	Smoking history	Smoker	0.38170
6	Brain metastasis	Yes	0.46250
4	Histology	Squamous	0.46970
15	LIPI score	Low risk	0.50060
10	PD-L1 TPS	$\geq 50\%$	0.51620
17	LIPI score	High risk	0.55690
8	PD-L1 TPS	$< 1\%$	0.61690
3	ECOG PS	0	0.65050
14	LDH	1 ULN	0.67330
5	Stage	IIIC	0.76500

7 Diagnostics

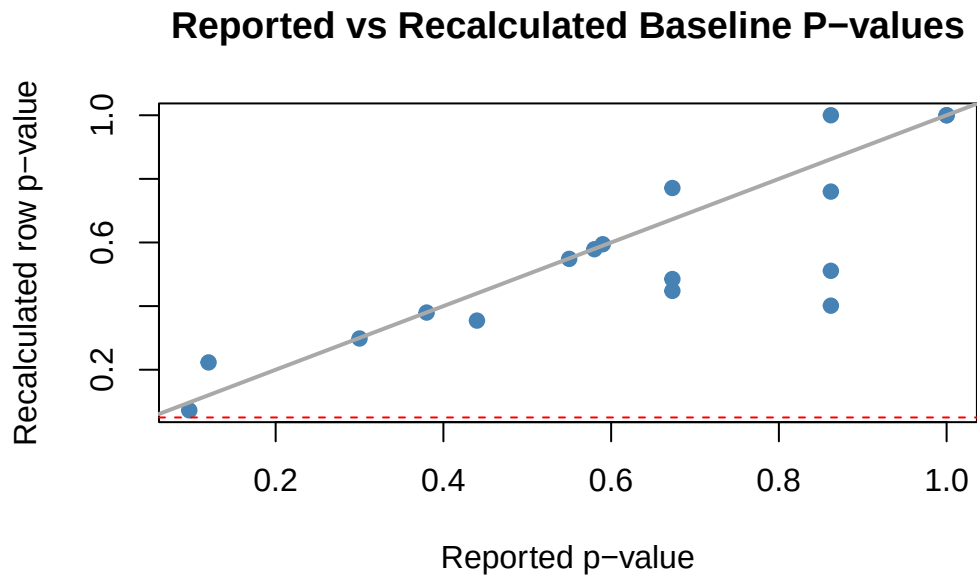
7.1 Distribution of recalculated row p-values

```
hist(  
  row_level$row_chisq_p,  
  breaks = 10,  
  col = "gray80",  
  border = "white",  
  main = "Row-level Recalculated P-values",  
  xlab = "row_chisq_p"  
)  
abline(v = 0.05, col = "red", lwd = 2, lty = 2)
```



7.2 Reported vs recalculated row p-values

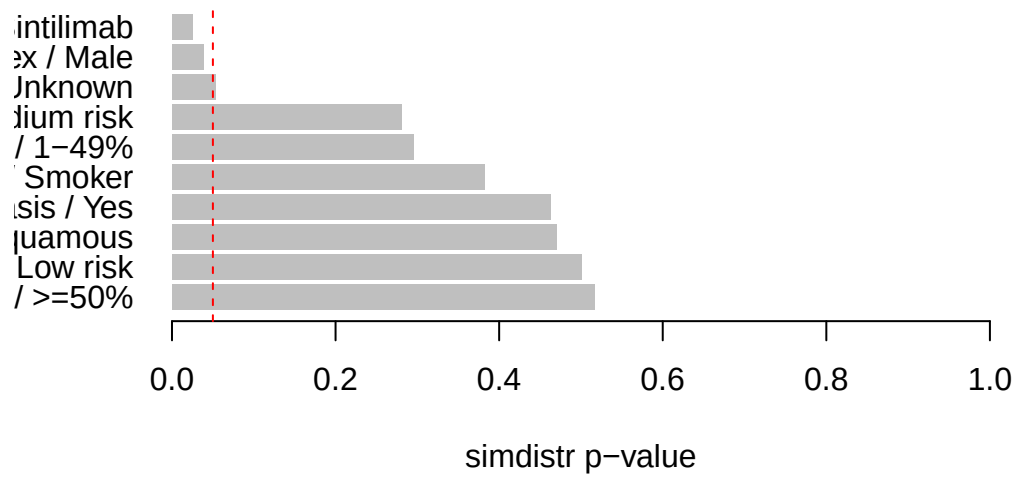
```
plot(  
  row_level$reported_p,  
  row_level$row_chisq_p,  
  pch = 19,  
  col = "steelblue",  
  xlab = "Reported p-value",  
  ylab = "Recalculated row p-value",  
  main = "Reported vs Recalculated Baseline P-values"  
)  
abline(0, 1, col = "darkgray", lwd = 2)  
abline(h = 0.05, col = "red", lty = 2)  
abline(v = 0.05, col = "red", lty = 2)
```



7.3 Lowest simdistr variable-level p-values

```
plot_df <- head(sim_by_p, 10)
barplot(
  height = rev(plot_df$simdistr_pvalue),
  names.arg = rev(paste(plot_df$variable, plot_df$level, sep = " / ")),
  horiz = TRUE,
  las = 1,
  col = "gray75",
  border = NA,
  main = "Lowest simdistr Variable-level P-values",
  xlab = "simdistr p-value",
  xlim = c(0, 1)
)
abline(v = 0.05, col = "red", lty = 2)
```

Lowest simdistr Variable-level P-values



8 Interpretation Notes

- This is a baseline-balance forensic screen, not a causal efficacy analysis.
- Row-level p-values depend on one-vs-rest encodings for multi-level variables.
- Reported manuscript p-values may use test choices that differ from recalculation methods.
- Joint p-value methods are sensitive to correlation among baseline variables.